



JUNE 30, 2025

Chemical Engineering Faculty Retreat Undergraduate Program

Alan Zdunek



UNIVERSITY OF
ILLINOIS CHICAGO



UIC Chemical Engineering Undergraduate Program

- EAB & Curriculum Update
- ABET preparation
- Recruitment Best Practices
- 2025-2026 plans



External Advisory Board (EAB)

Recently Stepped down:

Mr. Michael H. Goluszka, LA-CO
Industries

Dr. Diane Graziano, Argonne
National Lab

Dr. Michael Seacrist, MEMC

SEPTEMBER MEETING BEING PLANNED

CONTINUING MEMBERS:

Tarun Chandra, Managing Principal, EmpiriQA LLC

Soundar Kumaran, Sr. Research Engineer, Cabot Corp.

Dennis O'Brien, Consultant, Jacobs Engineering (Ret)

Anil Oroskar, CTO, Orochem Technologies

Charles Damianides, Executive Director, US DOE,
Center for Sustainable Fuels & Chemicals

New members:

Saadet Acikgoz, R&D Director, Process Design,
Modeling, and Digital Technology, UOP/Honeywell

Shailendra Bordawekar, VP Synthetic Molecules CMC
R&D Development Sciences, Abbvie

Claus Daniel, Associate Lab Director, Advanced
Energy Technologies, Argonne National Laboratory





ChE Curriculum

Note: CHE 150 Climate Engineering is a General Education course offered to Non-ChE majors

Core Courses

Thermodynamics-Based Courses

ChE 201 – Introduction to thermodynamics
ChE 301 – Chemical Engineering Thermodynamics
ChE 321 – Chemical Reaction Engineering

Process-Oriented Courses

ChE 210 – Material & Energy Balances
ChE 341 – Chemical Process Control
ChE 396 – Senior Design I (Equipment Design)
ChE 397 – Senior Design II (Design teams)

Numerical Analysis

ChE 205- Numerical Methods in ChE

Engineering Education & Development

ENGR100 -Engineering Orientation (Engineering Success Program, Guaranteed Paid Internship Program)
ChE 499- Professional Development Seminar

Transport-Based Courses

ChE 311 – Transport Phenomena I (Fluid Mechanics)
ChE 312 – Transport Phenomena II (Heat & Mass Transfer)
ChE 313 – Transport Phenomena III (Separations)

Unit Operations Laboratory

ChE 381 – Chem Eng Lab I
ChE 382 – Chem Eng Lab II

Molecular Engineering Courses

CHE 230 Molecular Engineering in CHE
CHE 330 Polymer Science

ChE 400-level Elective
CHE 392:
Undergraduate
Research

Six Concentrations:

Biochemical Engineering

Energy & Environment

Process Automation

Nanotechnology

Polymers & Molecular Engineering

Entrepreneurship





400-level courses

New permanent
course: CHE 426

New CHE 494
courses

CHE 410 Transport Phenomena

CHE 413 Introduction to Flow in Porous Media

CHE 421 Combustion Engineering

CHE 422 Biochemical Engineering

CHE 423 Catalytic Reaction Engineering

CHE 425 Nanotechnology for Pharmaceutical Applications

★ **CHE 426 Biopharmaceutical Process Development and Manufacturing**

CHE 427 Entrepreneurship in Engineering

CHE 433 Process Simulation w/Aspen+

CHE 438 Computational Molecular Modeling

CHE 440 Non-Newtonian Fluids

CHE 441 Computer Applications in Chemical Engineering

CHE 451 Renewable Energy Technology

CHE 450 Air Pollution

CHE 453 Electrochemistry

CHE 454 Molecular & Macromolecular Engineering

CHE 455 Nanoscale Systems in Chemical Engineering

CHE 456. Fundamentals and Design of Microelectronics Processes

CHE 457 Colloidal and Interfacial Phenomena

CHE 494 Molecular Gastronomy and Engineering

CHE 494 Introduction Chemical Research Methods in Chemical Engineering

CHE 494 Battery and Fuel Cell Development

CHE 494 Clean Energy & the Electric Grid.....



Curriculum Improvement: CHE 101 Course Revision

Approved by the COE
Education Policy Committee
(EPC), April 2025

CHE 101: Introduction to Chemical Engineering and Sustainability

Overview of the principles and unit operations of chemical engineering as practiced in traditional industries and in sustainable technologies.

Chemical engineering/unit operations, ChE traditional processes, case studies

Five sustainable themes (modules):
Energy, Health, Food, Air, Water

Students will explore the challenges in these areas and the role of chemical engineering in developing sustainable products and processes.

Modules will include a guest speaker (faculty or external)



Curriculum Improvement: Program Revision

ADD CHE 101 TO CORE BS REQUIREMENT (+3 CREDIT HOURS)

Initial plan: Remove CHEM 234 Org Chem II

Analysis:

- A review of chemical engineering programs shows two organic chemistry courses are required by Purdue, Northwestern, UW-Madison, UMich-Ann Arbor. UIUC requires Org I, with selective option (Org II or Biochemistry)
- External advisory board: Keep Organic Chem II, important topics covered
- UIC ChE student evaluation: Remove CHEM 234, not interesting, not needed.
- ABET Math/Science course requirements not an issue (minimum 30 credit hours, we have 47 credit hours)



Curriculum Improvement: Program Revisions

ADDING CHE 101(+3 CREDIT HOURS)

POSSIBLE SOLUTIONS:

- Remove CHEM 234
- Remove CHE 230
- Remove CHEM 342 Physical Chemistry I
- Remove ECE 210 Electric Circuits
- Make CHE 230 and CME 260 a Selective Option
 - Mat Sci requirement of other ChE programs varies: some require it (UMich, NU), some have it as a selective option (UW-Madison), some do not require it in core curriculum (UIUC, Purdue)



ABET Accreditation Review- Fall 2026

Timeline Faculty Input Course Assessment Example

TIMELINE

- **Summer 2026:**
 - Gather information- Box folder
 - Course information- Update Learning outcomes, performance indicators, upload student output, ABET syllabi
 - Faculty course rubric assessments
- **Fall 2026**
 - Gather information for Self-Study Report
 - Perform assessment analyses
 - Faculty information
 - Continue course rubric assessments-faculty
 - Continue updating course information-faculty
- **Spring 2026**
 - Gather Department, College, University, non-ChE course information
 - Self-study report deadline ~ June 2026
- **Summer 2026**
 - Modify and update ABET Review presentation system
 - Additional updates to Self-study report
- **Fall 2026**
 - Finalize presentation system
 - Plan for ABET reviewer visit
 - ABET review; reviewer and date TBD



ABET Accreditation Review- Fall 2026

Self-Study Report Due June 2026

Faculty
input

BACKGROUND INFORMATION	3
CRITERION 1. STUDENTS.....	7
CRITERION 2. PROGRAM EDUCATIONAL OBJECTIVES	23
CRITERION 3. STUDENT OUTCOMES	32
CRITERION 4. CONTINUOUS IMPROVEMENT.....	39
CRITERION 5. CURRICULUM.....	76
CRITERION 6. FACULTY	89
CRITERION 7. FACILITIES	101
CRITERION 8. INSTITUTIONAL SUPPORT	109
PROGRAM CRITERIA	115
Appendix A – Course Syllabi	117
Appendix B – Faculty Vitae	210
Appendix C – Equipment.....	242
Appendix D – Institutional Summary	245



ABET Accreditation Review- Fall 2026

Faculty Input

FACULTY INPUT FOR ABET REVIEW

- **Course information**
 - ABET course syllabi
 - Examples of student output
 - Course rubric assessment and evidence of continuous course improvement
- **Faculty information**
 - ABET CV's, qualifications, workload, photo
 - Awards, accomplishments,
 - Research and academic collaborations,
 - Research and academic initiatives



ABET Assessment Criterion 3 & 4

7 Student Learning Outcomes

**Students graduating
from the chemical
engineering program
at UIC will have:**

SO	Description
1	An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
2	An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
3	An ability to communicate effectively with a range of audiences.
4	An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.
5	An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.
6	An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.
7	An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

ABET Assessment

Student Learning Outcomes Mapped to ChE Courses

Course	(1) An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	(2) An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	(3) An ability to communicate effectively with a range of audiences.	(4) An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	(5) An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	(6) An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	(7) An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.
CHE 201	3	2	2	2	2	1	2
CHE 205	4	2	1	1	1	2	1
CHE 210	4	2	1	1	1	1	1
CHE 301	3	2	3	2	1	1	2
CHE 311	4	4	1	1	1	1	1
CHE 312	4	3	1	2	1	1	3
CHE 313	4	4	3	2	3	3	4
CHE 321	4	4	2	3	3	2	3
CHE 341	4	4	3	1	1	1	1
CHE 381	4	1	4	3	4	4	3
CHE 382	4	1	4	3	4	4	3
CHE 396	4	4	3	4	4	3	3
CHE 397	4	4	4	4	4	3	3
CHE 422	2	3	3	1	1	1	2
CHE 425	4	4	3	3	3	2	4
CHE 433	4	1	1	1	1	1	1
CHE 451	3	4	4	4	4	1	4
CHE 453	4	4	2	1	1	2	1
CHE 499	1	1	3	3	1	1	3

4 = topic is main focus of course, grade largely dependent on demonstrated knowledge;

3 = course grade partially dependent on demonstrated knowledge on this topic;

2 = topic introduced in course, course grade not dependent on demonstrated knowledge;

1 = topic not covered specifically in course.



ABET Assessment

Performance Indicators for SLO 1

Used to perform
course rubric
assessment on all
students in course

Performance Indicator (PI)	Level 3: Satisfactory	Level 2: Developing	Level 1: Unsatisfactory
1. Apply mathematical and scientific principles to formulate models and systems relevant to the subject	Able to successfully combines mathematical and/or scientific principles to formulate models and systems relevant to chemical engineering	Chooses a mathematical model or scientific principle that applies to an engineering problem, but has trouble in model development	Does not understand the connection between mathematical models and the system or process to be analyzed or designed
2. Solve engineering problems by using the concepts of algebra, calculus, differential equations and/or linear algebra	applies concepts of algebra, calculus, differential equations and/or linear algebra to solve chemical engineering problems	Shows nearly complete understanding of applications of calculus and/or linear algebra in problem-solving	Does not understand the application of algebra, calculus, differential equations and/or linear algebra in solving chemical engineering problems
3. Appropriate engineering interpretation of mathematical and scientific terms	Shows appropriate engineering interpretation of mathematical and scientific terms	Most mathematical terms are interpreted correctly	Mathematical terms are interpreted incorrectly or not at all
4. Translates academic theory into engineering applications	Translates academic theory into engineering applications and accepts limitations of mathematical models of physical reality	Some gaps in understanding the application of theory to the problem and expects theory to predict reality	Does not appear to grasp the connection between theory and the problem
5. Predict and defend problem outcomes	Can predict and defend problem outcomes	Occasionally predicts and defends problem outcomes	Is unable to predict or defend problem outcomes
6. The uses of appropriate resources needed to solve problems	Uses appropriate resources to locate information needed to solve problems	Uses limited resources to solve problems	Uses no resources to solve problems
7. Solutions creativity alternatives	Demonstrates creative synthesis of solution and creates new alternatives by combining knowledge and information	Demonstrates solution with integration of diverse concepts or derivation of useful relationships involving ideas covered in course concepts; however, no alternative solutions are generated	Demonstrates solutions implementing simple applications of one formula or equation with close analogies to class/lecture problems

Student Learning Outcome Course Rubric Example

ChE 205- Computational Methods in Chemical Engineering- Spring 2022

Student Learning Outcome 1

Rubric for assessment for Final exam, Problems, 1c, 2, 4, and 3a

RESULTS BY PERFORMANCE INDICATOR					
Rubric Item (PI)	1	2	4	6	
Mapped SLO	1	1	1	1	
Average	2.25	2.25	2.10	2.85	
Std Dev	0.79	0.64	0.79	0.37	

RESULTS BY STUDENT OUTCOME	
SO	1
Average Score	2.36
Standard dev.	0.29

Analysis:

Average scores were fairly uniform across Performance Indicators (PI) 1, 2, and 4. The strongest score was for Performance Indicator 6; Using appropriate resources to solve problems. A lower score was observed for PI 4, Translating academic theory into engineering applications. There was a fairly even distribution between satisfactory and developing scores for PI's 1, 2 and 4. This data suggests that most students were strong in (i) using appropriate resources needed to solve problems, were developing in (ii) applying mathematical and scientific principles to formulate models, and (iii) solving engineering problems, but were weak in (iv) translating academic theory into engineering applications.

Points of Improvement:

- There is a broad range of student skill level in programming and Excel. Most students are still developing these skills and need extra help. It would be beneficial to cover the basics of Excel and programming at the beginning of the course and provide resources on those topics.

SLO 1 Performance Indicator (PI)	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
1. Apply mathematical and scientific principles to formulate models and systems relevant to the subject	Able to successfully combines mathematical and/or scientific principles to formulate models and systems relevant to chemical engineering	Chooses a mathematical model or scientific principle that applies to an engineering problem, but has trouble in model development	Does not understand the connection between mathematical models and the system or process to be analyzed or designed



Student Outcome Assessment Summary

Basis:

Course Rubrics Senior Exam Senior EXPO

Summary of “S” and “D” attainment for all Student Outcomes:

3 = S = Satisfactory 2 = D = Developing 1 = U = Unsatisfactory

Percentage of students attaining a rating of Satisfactory for all the SO's

SO	2020-2021	2021-2022
1	78.4	73.0
2	72.0	86.0
3	80.3	81.2
4	82.6	84.8
5	72.7	82.9
6	75.1	71.5
7	71.1	85.4

Percentage of students attaining a rating of Satisfactory or Developing for all SO's

SO	2020-2021	2021-2022
1	94.6	90.4
2	97.1	95.3
3	95.9	97.7
4	98.8	96.7
5	97.9	94.8
6	93.9	90.1
7	97.1	98.6



Chemical Engineering Recruitment: Best Practices



FALL 2025: 66 INTENT TO ENROLL (ITE)

Compared to 43 ITE's in 2024, 48 ITE's in 2023

More students were admitted to our program:

196- 2025

148- 2024

130- 2023

WHAT WE DID THIS YEAR:

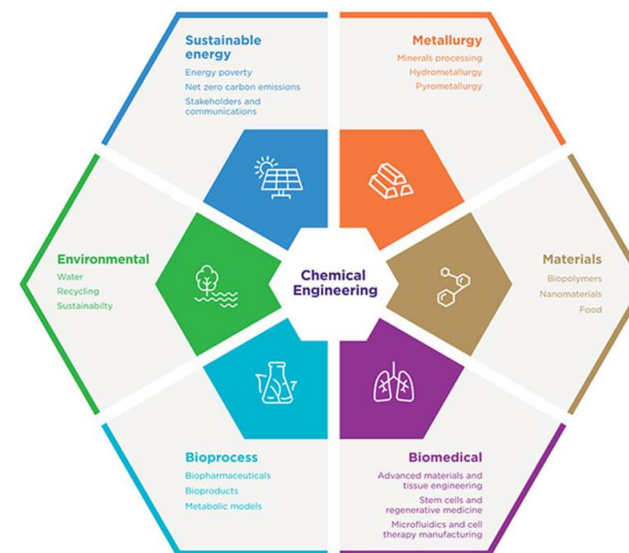
- **REFOCUSED MARKETING MESSAGE TO EMPHASIZE SUSTAINABILITY AND CHE DIVERSITY**
- **FOLLOW UP COMMUNICATION WITH ITE STUDENTS**
- **NEW UIC DISTRIBUTION LIST MAILING FORMAT (EMMA)**
- **CHE DEPARTMENT YIELD SESSION (MARCH)**
- **PHONE CALL CAMPAIGN TO ADMITTED STUDENTS**
- **CHE UG STUDENT INVOLVEMENT IN RECRUITMENT EVENTS**
- **HIGH SCHOOL ENGAGEMENT**
 - Northside College Prep (with COE team)
 - Woode STEM Academy (with COE team)
 - Leyden Township HS (Berry initiative)

Chemical Engineering Recruitment:

Sustainability/ ChE Diversity Message

Why it is Cool?

Combines science, math, and creativity
Solve global problems
Help people live healthier, better lives
Great career opportunities and variety



"Chemical engineers don't just work with chemicals—they work with ideas, innovation, and impact."

16

Unlock the Future with Chemical Engineering!

Imagine being at the forefront of groundbreaking innovations. Chemical engineers are transforming the world by creating renewable energy, providing clean water, developing life-saving drugs, designing eco-friendly products, and using AI and nanotechnology to revolutionize industries. Ready to make a real difference and change the world? Join us and become part of this amazing journey!

Chemical Engineering Recruitment: Mailing letter (EMMA)

**UNIVERSITY OF
ILLINOIS CHICAGO**



Choose Chemical Engineering- A Sustainable Major

Hello there,

Our chemical engineering students put on quite the show with their Senior Design EXPO projects this semester! From creating new bioethanol fuel through non-food biomass sources, using carbon-capture to produce methanol, and redesigning plastic production to keep it out of landfills and waterways, the students tackled global problems with sustainable ideas and inventions. Learn more about the projects here: <https://lnkd.in/g/jfQC4zS>

[hashtag#UIC](#) [hashtag#UICCollegeofEngineering](#) [hashtag#ChemicalEngineering](#)
[hashtag#SeniorDesignExpo](#)

Subscribe to our email list.



Chemical Engineering Recruitment: Improvement

- **Attend ENGR100 Undeclared Major sections**
 - ChE UG students or recent alumni
- **Keep communicating to ITE's in Summer**
 - Get all ITE's to register
- **Require a fee to enroll (ITE)**
- **More collaborations with COE recruitment team**
 - HS events, tours, Yield session
- **CHE Faculty participation at recruitment events**
- **Increase High school engagement**
- **Other ideas?**



Overview of 2025-2026 Plans

- **ABET Preparation: Summer 2025----**
- **EAB Meeting- mid/late September**
- **Alumni Event: October-November**
- **Curriculum Proposal: Fall 2026**

Recruitment events: Need faculty participation

- SparkFest 2025 (Open House) – September 20th
- WIE Scholarship Brunch- February 2026
- Department Yield Session- March 2026
- IGNITE Event- April 2026
- High School presentations- TBD
- COE recruitment team collaboration



CHEM 232 and CHEM 234 Topics

CHEM 232

- **Electrons, Bonds, and Molecular Properties**
- **Molecular Representations**
- **Acids and Bases**
- **Alkanes and Cycloalkanes**
- **Stereoisomerism**
- **Chemical Reactivity and Mechanisms**
- **Substitution Reactions**
- **Alkenes: Structure and Preparation via Elimination Reactions**
- **Addition Reactions of Alkenes**
- **Alkynes**
- **Radical Reactions**
- **Synthesis**
- **Alcohols and Phenols**
- **Ethers and Epoxides; Thiols and Sulfides**
- **Infrared Spectroscopy and Mass Spectrometry**

CHEM 234

- **Ethers and Epoxides; Thiols and Sulfides**
- **Infrared Spectroscopy and Mass Spectrometry**
- **Nuclear Magnetic Resonance Spectroscopy**
- **Conjugated Pi Systems and Pericyclic Reactions**
- **Aromatic Compounds**
- **Aromatic Substitution Reactions**
- **Aldehydes and Ketones**
- **Carboxylic Acids and Their Derivatives**
- **Alpha Carbon Chemistry: Enols and Enolates**
- **Amines**
- **Carbohydrates**
- **Amino Acids, Peptides, and Proteins**
- **Lipids**
- **Synthetic Polymers**



CHE 230 and CME 260 Topics

CHE 230

Topic
Introduction to Molecular Systems in Chemical Engineering
Fundamentals of Molecular Design
Material Properties
Polymer Science
Synthesis and Characterization of Nanomaterials
Self-Assembled Materials: Driving Forces and Characterization
Engineering Functional Materials

CME 260

- Introduction, Atomic Bonding
- Crystallography
- Crystal Defects and Diffusion
- Mechanical Properties of Metals
- Phase Diagrams
- Phase transformations
- Processing of Metals
- Ceramics
- Polymers and Composites
- Electrical and Magnetic Materials





Number of students in each concentration

No students in Entrepreneurship- how can we promote it?

